White Box Testing Report

2024101087

Moneypoly — Part 1 In-Depth Analysis

`Current execution: 120 passed in 0.07s  |  113 test functions listed`

Scoring rule (per iteration): score = ((passed + xpassed) / (passed + failed + errors + xfailed + xpassed)) × 10

Section A — Iteration-wise Detailed Report (6 Iterations)

Method: For each iteration commit, objective, diff/files changed, production files touched, test functions added, and pytest terminal output were collected.

ITERATION 1

Commit: 6f3e59d **Score: 0.0/10**

Objective	Knock out the easy pylint noise first
Feature Changed	Repository-wide code hygiene and lint cleanup across core MoneyPoly modules.
Diff Impact	**+123 / -42**

Files Changed	`moneypoly/moneypoly/main.py`
	`moneypoly/moneypoly/moneypoly/bank.py`
	`moneypoly/moneypoly/moneypoly/board.py`
	`moneypoly/moneypoly/moneypoly/cards.py`
	`moneypoly/moneypoly/moneypoly/config.py`
	`moneypoly/moneypoly/moneypoly/dice.py`
	`moneypoly/moneypoly/moneypoly/game.py`
	`moneypoly/moneypoly/moneypoly/player.py`
	`moneypoly/moneypoly/moneypoly/property.py`
	`moneypoly/moneypoly/moneypoly/ui.py`

Prod Files	10 production file(s) modified
Test Functions Added	`None (no new test_ function added in this commit)`
Deep Analysis	• Established a baseline quality floor by removing broad lint/style debt across core modules. • This increased semantic clarity of later commits and reduced review noise.
Terminal Output	`no tests ran in 0.00s` `ERROR: file or directory not found: whitebox/tests`
Score	**N/A since test are not run for this**

ITERATION 2

Commit: 3e34f38 **Score: 0.0/10**

Objective	Refactor the core models until pylint is clean
Feature Changed	Core model refactor in game.py, player.py, and property.py for cleaner logic boundaries.
Diff Impact	**+73 / -49**

Files Changed	`moneypoly/moneypoly/moneypoly/game.py`
	`moneypoly/moneypoly/moneypoly/player.py`
	`moneypoly/moneypoly/moneypoly/property.py`

Prod Files	3 production file(s) modified
Test Functions Added	`None (no new test_ function added in this commit)`
Deep Analysis	• Refactored central game-domain models where branch and state complexity are highest. • Improved separations of concern and made future defect fixes easier to target.
Terminal Output	`no tests ran in 0.00s` `ERROR: file or directory not found: whitebox/tests`
Score	**N/A for this test**

ITERATION 3

Commit: fb6d824 **Score: 9.37/10**

Objective	Expand helper and CLI decision coverage
Feature Changed	Added white-box coverage for helper and CLI decision paths.
Diff Impact	**+284 / -0**
Files Changed	`whitebox/tests/test_helpers_and_cli.py`
Prod Files	None (test-only iteration)

Test Functions Added	`test_bank_zero_payout_and_insufficient_payout`
	`test_bank_give_loan_ignores_non_positive_amounts`
	`test_bank_summary_and_repr`
	`test_board_reports_blank_and_special_tiles`
	`test_board_purchasable_checks_cover_none_owned_and_mortgaged`
	`test_board_repr_reflects_owned_property_count`
	`test_card_deck_peek_len_repr_and_reshuffle`
	`test_dice_reset_describe_and_repr`
	`test_player_validation_and_display_helpers`
	`test_property_helpers_cover_mortgage_availability_and_repr`
	`test_property_group_helpers_cover_owner_counts_size_and_repr`
	`test_ui_print_helpers_cover_player_board_and_standings`
	`test_ui_input_helpers_cover_fallbacks_and_negative_confirmation`
	`test_get_player_names_trims_and_filters_empty_entries`
	`test_main_runs_the_game_when_setup_succeeds`
	`test_main_handles_keyboard_interrupt`
	`Test_main_handles_setup_value_errors`
	`Test_game_rejects_invalid_player_counts`
	`Test_move_collects_salary_when_passing_go`
	`Test_find_winner_uses_highest_net_worth`
	`Test_board_exposes_railroads_as_real_properties`
	…

Deep Analysis	• Shifted quality focus to helper/CLI decision paths that are commonly under-tested. • Improved resilience against invalid/edge user input and startup flow issues.

Terminal Output	`.....`
	`[100%]`
	`59 passed in 0.16s`
	`4 failed in 0.14 s`

Score	**9.37/10**

ITERATION 4

Commit: 04df64c **Score: 9.35/10**

Objective	Cover the remaining game decision paths
Feature Changed	Added white-box coverage for remaining game decision branches.
Diff Impact	**+476 / -0**
Files Changed	`whitebox/tests/test_game_decision_paths.py`
Prod Files	None (test-only iteration)

Test Functions Added	`test_play_turn_advances_after_a_regular_roll`
	`test_play_turn_keeps_the_same_player_after_doubles`
	`test_play_turn_sends_the_player_to_jail_after_three_doubles`
	`test_resolve_current_tile_leaves_non_action_tiles_alone`
	`test_resolve_current_tile_draws_a_chance_card`
	`test_resolve_current_tile_draws_a_community_chest_card`
	`test_resolve_current_tile_ignores_missing_property_objects`
	`test_handle_property_tile_can_start_an_auction`
	`test_handle_property_tile_skips_when_the_player_passes`
	`test_handle_property_tile_notices_when_the_player_already_owns_it`
	`test_buy_property_reports_insufficient_funds`
	`test_pay_rent_ignores_unowned_property`
	`test_mortgage_property_rejects_wrong_owner_and_repeated_mortgages`
	`test_unmortgage_property_rejects_wrong_owner_and_non_mortgaged_spaces`
	`test_trade_rejects_wrong_owner_and_insufficient_buyer_balance`
	`test_auction_property_leaves_it_unowned_when_everyone_passes`
	`test_handle_jail_turn_uses_a_jail_free_card`
	`test_handle_jail_turn_waits_when_the_player_declines_to_leave`
	`test_handle_jail_turn_forces_release_after_three_turns`
	`test_apply_card_handles_none_and_unknown_actions`
	`test_apply_card_pay_and_jail_paths`
	`test_collect_from_other_players_skips_people_who_cannot_afford_it`
	`test_check_bankruptcy_leaves_solvent_players_in_the_game`
	`test_find_winner_returns_none_for_an_empty_game`
	`test_run_prints_a_winner_when_one_player_remains`
	`test_run_handles_an_empty_player_list`
	`test_interactive_menu_dispatches_all_visible_options`
	`test_interactive_menu_skips_non_positive_loan_requests`
	`test_menu_mortgage_covers_empty_and_valid_selection`
	`test_menu_unmortgage_covers_empty_and_valid_selection`
	`Test_menu_trade_covers_missing_players_invalid_choices_and_success`
	`Test_pay_rent_transfers_cash_to_owner`
	`Test_unmortgage_fails_without_changing_state_when_cash_is_short`
	`Test_give_loan_reduces_bank_reserves_and_tracks_it`
	`Test_dice_rolls_use_the_full_six_sided_range`
	`Test_bankruptcy_does_not_skip_the_next_player`
	`Test_move_to_card_resolves_tax_spaces`
	`Test_play_turn_opens_the_pre_roll_menu`
	…

Deep Analysis	• Expanded decision-path tests in the game loop to reveal latent branch defects. • Increased confidence that non-happy-path turn behaviors are explicitly validated.

Terminal Output	`.....`
	`[ 72%]`
	`  .....`
	`[100%]`
	`  100 passed in 0.24s`
	`  7 failed in 0.12s`

Score	**9.35/10**

ITERATION 5

Commit: fd03b08 **Score: 9.5/10**

Objective	Tighten coverage around the last branch edges
Feature Changed	Hardened edge-branch coverage in decision and CLI/helper tests.
Diff Impact	**+245 / -0**
Files Changed	`whitebox/tests/test_game_decision_paths.py`

	`whitebox/tests/test_helpers_and_cli.py`
Prod Files	None (test-only iteration)

Test Functions Added	`test_play_turn_processes_jailed_players_and_then_advances`
	`test_play_turn_handles_jailed_players_removed_during_jail_logic`
	`test_handle_property_tile_charges_rent_when_another_player_owns_it`
	`test_resolve_current_tile_handles_standard_properties`
	`test_unmortgage_property_succeeds_for_the_owner`
	`test_handle_jail_turn_can_decline_the_card_and_pay_instead`
	`test_apply_card_birthday_and_move_to_go_cover_remaining_handlers`
	`test_check_bankruptcy_releases_properties_and_resets_index`
	`test_check_bankruptcy_tolerates_players_already_removed_from_the_game`
	`test_run_executes_the_loop_and_prints_standings`
	`test_menu_mortgage_ignores_invalid_selection`
	`test_menu_unmortgage_ignores_invalid_selection`
	`test_dice_roll_increments_the_doubles_streak`
	`test_ui_print_player_card_handles_players_without_properties`
	`Test_main_module_runs_from_the_script_entry_point`
	`Test_player_with_zero_cash_and_assets_is_not_eliminated`
	`Test_doubles_streak_resets_when_turn_passes_to_the_next_player`
	`Test_buy_property_rejects_mortgaged_unowned_property`
	`Test_play_turn_sends_the_player_to_jail_after_three_doubles`
	`Test_interactive_menu_handles_unaffordable_emergency_loans`
	`test_main_handles_eof_during_initial_player_input`

Deep Analysis	• Tightened edge assertions, reducing ambiguity when failures occur. • This iteration improved failure localization and debugging speed.

Terminal Output	`.....`
	`[ 62%]`
	`  .....`
	`[100%]`
	`  115 passed in 0.28s`
	`  6 failed in 0.23 s`

Score	**9.5/10**

ITERATION 6

Commit: 1103612 **Score: 10.0/10**

Objective	Close the last branch gaps for full coverage
Feature Changed	Closed residual branch gaps to reach high confidence coverage for Part 1 scope.
Diff Impact	**+18 / -2**
Files Changed	`whitebox/tests/test_game_decision_paths.py` `whitebox/tests/test_helpers_and_cli.py`
Prod Files	None (test-only iteration)
Test Functions Added	`test_interactive_menu_ignores_invalid_choices_and_loops`
Deep Analysis	• Closed residual branch gaps and stabilized the Part 1 white-box test surface. • Marks the point where coverage confidence reached submission-level maturity.

Terminal Output	`.....`
	`[ 62%]`
	`  .....`
	`[100%]`
	`  116 passed in 0.28s`

Score	**10.0/10**

Final Outcome: 10/10 after completing Iteration 6.

Section B — 17 Errors: Test, Need, Logical Issue, Fix Made

All 17 errors listed below were historical failures. Each was identified through white-box branch analysis, fixed in a dedicated commit, and now passes in the current run.

Error #	White-box Test Case	Why This Test Was Needed	Issue Found → Fix Made (files updated)
1	**Error 1: Stop invalid games before the first turn** Commit: c8d6d92 test_game_rejects_invalid_player_counts	Needed to validate a distinct white-box branch/edge path for correctness and regression safety.	**Issue:** Game setup allowed invalid player count states to proceed; this is a guard failure at the start of execution and can invalidate all downstream behavior. Change volume: +44/-1 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/game.py pytest.ini whitebox/tests/conftest.py whitebox/tests/test_setup.py
2	**Error 2: Pay players when they pass Go, not just land on it** Commit: 1417629 test_move_collects_salary_when_passing_go	Needed to verify pass-Go salary behavior and movement economy rules.	**Issue:** Economy logic underpaid players by awarding salary only on landing instead of also on passing Go. Change volume: +40/-3 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/player.py whitebox/tests/test_player_movement.py
3	**Error 3: Score assets correctly before picking a winner** Commit: 6f063d1 test_find_winner_uses_highest_net_worth	Needed to ensure winner/asset valuation logic is accurate and deterministic.	**Issue:** Winner selection used incomplete or incorrect net-worth computation, causing wrong game outcomes. Change volume: +66/-3 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/game.py moneypoly/moneypoly/moneypoly/player.py moneypoly/moneypoly/moneypoly/property.py whitebox/tests/test_assets_and_winner.py
4	**Error 4: Make railroad spaces and edge-case buys work** Commit: 6e3a958 test_board_exposes_railroads_as_real_properties	Needed to validate board/property invariants, ownership, and purchase rules.	**Issue:** Board modeling/purchase edge logic mishandled railroads or edge-case buy paths. Change volume: +45/-1 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/board.py moneypoly/moneypoly/moneypoly/game.py whitebox/tests/test_board_spaces.py
5	**Error 5: Send rent and trade cash to the right player** Commit: fbacc2e test_pay_rent_transfers_cash_to_owner	Needed to confirm money transfers are credited/debited to the correct players.	**Issue:** Transfer paths (rent/trade) sent money to wrong destination player or skipped correct credit/debit behavior. Change volume: +67/-0 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/game.py whitebox/tests/test_transfers.py
6	**Error 6: Only release jail and mortgages after real payment** Commit: 0f87f2a test_unmortgage_fails_without_changing_state_when_cash_is_short	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.	**Issue:** Jail/mortgage release transitions occurred without validating real payment completion. Change volume: +81/-11 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/game.py moneypoly/moneypoly/moneypoly/property.py whitebox/tests/test_jail_and_mortgage.py
7	**Error 7: Run loans and mortgages through the bank ledger** Commit: 51043d8	Needed to verify loan handling and bank ledger consistency.	**Issue:** Loan/mortgage accounting bypassed correct bank ledger behavior, risking reserve inconsistencies. Change volume: +51/-5 | Post-fix: Historical Fail → Current Pass **Files:**

Error #	White-box Test Case	Why This Test Was Needed	Issue Found → Fix Made (files updated)
	test_give_loan_reduces_bank_reserves_and_tracks_it		moneypoly/moneypoly/moneypoly/bank.py moneypoly/moneypoly/moneypoly/game.py whitebox/tests/test_banking.py
8	**Error 8: Fix the dice bounds and safe deck helpers** Commit: ab9d91c test_dice_rolls_use_the_full_six_sided_range	Needed to ensure helper randomness/deck operations stay within valid bounds.	**Issue:** Dice/deck helpers had bounds/safety issues causing invalid random outcomes or unsafe helper behavior. Change volume: +50/-2 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/cards.py moneypoly/moneypoly/moneypoly/dice.py whitebox/tests/test_dice_and_decks.py
9	**Error 9: Keep turn order stable after a bankruptcy** Commit: e15965a test_bankruptcy_does_not_skip_the_next_player	Needed to verify bankruptcy transitions are correct and players are not wrongly eliminated.	**Issue:** Turn index/order corrupted after bankruptcy, causing skipped or repeated players. Change volume: +28/-1 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/game.py whitebox/tests/test_turn_order.py
10	**Error 10: Resolve every destination when a card moves a player** Commit: d7ba03b test_move_to_card_resolves_tax_spaces	Needed to verify card destination/action resolution and its side effects.	**Issue:** Card-driven movement failed to resolve all destination types correctly. Change volume: +55/-5 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/game.py whitebox/tests/test_card_destinations.py
11	**Error 11: Actually show the pre-roll menu during each turn** Commit: a7e762b test_play_turn_opens_the_pre_roll_menu	Needed to validate branch-heavy turn flow and state transition correctness.	**Issue:** Turn menu was not reliably shown before roll, creating control-flow inconsistency. Change volume: +22/-0 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/game.py whitebox/tests/test_turn_menu.py
12	**Error 12: Stop bankrupting players who still have recoverable assets** Commit: a80c5c9 test_player_with_zero_cash_and_assets_is_not_eliminated	Needed to ensure winner/asset valuation logic is accurate and deterministic.	**Issue:** Players with recoverable assets were declared bankrupt prematurely. Change volume: +45/-2 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/player.py whitebox/tests/test_bankruptcy_logic.py
13	**Error 13: Reset doubles streak when the turn passes** Commit: 5858f1a test_doubles_streak_resets_when_turn_passes_to_the_next_player	Needed to validate branch-heavy turn flow and state transition correctness.	**Issue:** Doubles streak state leaked across turn boundaries instead of resetting correctly. Change volume: +29/-1 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/game.py whitebox/tests/test_doubles_logic.py whitebox/tests/test_turn_order.py
14	**Error 14: Block direct purchases of unavailable properties** Commit: bd16d34 test_buy_property_rejects_mortgaged_unowned_property	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.	**Issue:** Direct purchase allowed unavailable/invalid property states, violating acquisition guards. Change volume: +32/-0 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/game.py whitebox/tests/test_purchase_guards.py

Error #	White-box Test Case	Why This Test Was Needed	Issue Found → Fix Made (files updated)
15	**Error 15: End turns after doubles send a player to jail** Commit: 235b9f0 test_play_turn_sends_the_player_to_jail_after_three_doubles	Needed to verify jail entry/release/payment paths and related state changes.	**Issue:** Turn continuation logic after jail-by-doubles was wrong (turn should terminate correctly). Change volume: +58/-0 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/game.py whitebox/tests/test_game_decision_paths.py
16	**Error 16: Reject oversized emergency loans without crashing** Commit: 466fd3d test_interactive_menu_handles_unaffordable_emergency_loans	Needed to validate branch-heavy turn flow and state transition correctness.	**Issue:** Emergency loan path accepted oversized values or crashed instead of rejecting safely. Change volume: +24/-1 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/moneypoly/game.py whitebox/tests/test_game_decision_paths.py
17	**Error 17: Handle startup EOF without a traceback** Commit: 31506bf test_main_handles_eof_during_initial_player_input	Needed to validate startup and CLI robustness under invalid or EOF inputs.	**Issue:** Startup EOF input path produced traceback instead of graceful handling. Change volume: +12/-2 | Post-fix: Historical Fail → Current Pass **Files:** moneypoly/moneypoly/main.py whitebox/tests/test_helpers_and_cli.py

Section C — Complete White-Box Test Case List (113 Tests)

All 113 tests listed below currently pass. Each was written to cover a specific branch, state invariant, or business rule identified during white-box analysis.

#	Test File	Test Function	Why Needed
001	test_assets_and_winner.py	`test_full_group_ownership_requires_every_property`	Needed to validate board/property invariants, ownership, and purchase rules.
002	test_assets_and_winner.py	`test_net_worth_counts_owned_property_value`	Needed to validate board/property invariants, ownership, and purchase rules.
003	test_assets_and_winner.py	`test_mortgaged_property_counts_at_mortgage_value`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
004	test_assets_and_winner.py	`test_find_winner_uses_highest_net_worth`	Needed to ensure winner/asset valuation logic is accurate and deterministic.
005	test_banking.py	`test_bank_collect_rejects_negative_amounts`	Needed to verify loan handling and bank ledger consistency.
006	test_banking.py	`test_give_loan_reduces_bank_reserves_and_tracks_it`	Needed to verify loan handling and bank ledger consistency.
007	test_banking.py	`test_mortgage_property_uses_bank_funds_for_the_payout`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
008	test_bankruptcy_logic.py	`test_player_with_zero_cash_and_assets_is_not_eliminated`	Needed to ensure winner/asset valuation logic is accurate and deterministic.
009	test_bankruptcy_logic.py	`test_player_with_negative_cash_and_assets_is_not_eliminated`	Needed to ensure winner/asset valuation logic is accurate and deterministic.
010	test_bankruptcy_logic.py	`test_player_with_negative_total_net_worth_is_eliminated`	Needed to ensure winner/asset valuation logic is accurate and deterministic.
011	test_board_spaces.py	`test_board_exposes_railroads_as_real_properties`	Needed to validate board/property invariants, ownership, and purchase rules.
012	test_board_spaces.py	`test_landing_on_railroad_uses_property_flow`	Needed to validate board/property invariants, ownership, and purchase rules.
013	test_board_spaces.py	`test_buy_property_allows_exact_balance_purchase`	Needed to validate board/property invariants, ownership, and purchase rules.
014	test_branch_coverage.py	`test_move_and_resolve_income_tax_collects_for_the_bank`	Needed to verify loan handling and bank ledger consistency.
015	test_branch_coverage.py	`test_move_and_resolve_luxury_tax_collects_for_the_bank`	Needed to verify loan handling and bank ledger consistency.
016	test_branch_coverage.py	`test_move_and_resolve_go_to_jail_square`	Needed to verify jail entry/release/payment paths and related state changes.
017	test_branch_coverage.py	`test_collect_card_pays_out_from_the_bank`	Needed to verify card destination/action resolution and its side effects.
018	test_branch_coverage.py	`test_jail_free_card_increases_the_player_inventory`	Needed to verify jail entry/release/payment paths and related state changes.

#	Test File	Test Function	Why Needed
019	test_branch_coverage.py	`test_auction_property_accepts_the_highest_valid_bid`	Needed to cover auction branch behavior and bid resolution rules.
020	test_branch_coverage.py	`test_board_helpers_track_owned_and_unowned_properties`	Needed to validate board/property invariants, ownership, and purchase rules.
021	test_branch_coverage.py	`test_safe_int_input_returns_the_default_on_invalid_text`	Needed to validate startup and CLI robustness under invalid or EOF inputs.
022	test_branch_coverage.py	`test_confirm_treats_lowercase_y_as_yes`	Needed to validate a distinct white-box branch/edge path for correctness and regression safety.
023	test_card_destinations.py	`test_move_to_card_resolves_railroad_purchase`	Needed to verify card destination/action resolution and its side effects.
024	test_card_destinations.py	`test_move_to_card_resolves_tax_spaces`	Needed to verify card destination/action resolution and its side effects.
025	test_card_destinations.py	`test_move_to_card_resolves_go_to_jail_spaces`	Needed to verify jail entry/release/payment paths and related state changes.
026	test_dice_and_decks.py	`test_dice_rolls_use_the_full_six_sided_range`	Needed to ensure helper randomness/deck operations stay within valid bounds.
027	test_dice_and_decks.py	`test_empty_card_deck_helpers_are_safe`	Needed to verify card destination/action resolution and its side effects.
028	test_dice_and_decks.py	`test_cards_remaining_wraps_after_the_deck_cycles`	Needed to verify card destination/action resolution and its side effects.
029	test_doubles_logic.py	`test_doubles_streak_resets_when_turn_passes_to_the_next_player`	Needed to validate branch-heavy turn flow and state transition correctness.
030	test_game_decision_paths.py	`test_play_turn_advances_after_a_regular_roll`	Needed to validate branch-heavy turn flow and state transition correctness.
031	test_game_decision_paths.py	`test_play_turn_keeps_the_same_player_after_doubles`	Needed to validate branch-heavy turn flow and state transition correctness.
032	test_game_decision_paths.py	`test_play_turn_sends_the_player_to_jail_after_three_doubles`	Needed to verify jail entry/release/payment paths and related state changes.
033	test_game_decision_paths.py	`test_play_turn_does_not_grant_an_extra_roll_after_doubles_land_on_go_to_jail`	Needed to verify jail entry/release/payment paths and related state changes.
034	test_game_decision_paths.py	`test_play_turn_does_not_grant_an_extra_roll_after_a_doubles_jail_card`	Needed to verify jail entry/release/payment paths and related state changes.
035	test_game_decision_paths.py	`test_play_turn_processes_jailed_players_and_then_advances`	Needed to verify jail entry/release/payment paths and related state changes.
036	test_game_decision_paths.py	`test_play_turn_handles_jailed_players_removed_during_jail_logic`	Needed to verify jail entry/release/payment paths and related state changes.
037	test_game_decision_paths.py	`test_resolve_current_tile_leaves_non_action_tiles_alone`	Needed to confirm money transfers are credited/debited to the correct players.

#	Test File	Test Function	Why Needed
038	test_game_decision_paths.py	`test_resolve_current_tile_draws_a_chance_card`	Needed to confirm money transfers are credited/debited to the correct players.
039	test_game_decision_paths.py	`test_resolve_current_tile_draws_a_community_chest_card`	Needed to confirm money transfers are credited/debited to the correct players.
040	test_game_decision_paths.py	`test_resolve_current_tile_ignores_missing_property_objects`	Needed to confirm money transfers are credited/debited to the correct players.
041	test_game_decision_paths.py	`test_handle_property_tile_can_start_an_auction`	Needed to cover auction branch behavior and bid resolution rules.
042	test_game_decision_paths.py	`test_handle_property_tile_skips_when_the_player_passes`	Needed to validate board/property invariants, ownership, and purchase rules.
043	test_game_decision_paths.py	`test_handle_property_tile_notices_when_the_player_already_owns_it`	Needed to validate board/property invariants, ownership, and purchase rules.
044	test_game_decision_paths.py	`test_handle_property_tile_charges_rent_when_another_player_owns_it`	Needed to confirm money transfers are credited/debited to the correct players.
045	test_game_decision_paths.py	`test_buy_property_reports_insufficient_funds`	Needed to validate board/property invariants, ownership, and purchase rules.
046	test_game_decision_paths.py	`test_resolve_current_tile_handles_standard_properties`	Needed to confirm money transfers are credited/debited to the correct players.
047	test_game_decision_paths.py	`test_pay_rent_ignores_unowned_property`	Needed to confirm money transfers are credited/debited to the correct players.
048	test_game_decision_paths.py	`test_mortgage_property_rejects_wrong_owner_and_repeated_mortgages`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
049	test_game_decision_paths.py	`test_unmortgage_property_rejects_wrong_owner_and_non_mortgaged_spaces`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
050	test_game_decision_paths.py	`test_unmortgage_property_succeeds_for_the_owner`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
051	test_game_decision_paths.py	`test_trade_rejects_wrong_owner_and_insufficient_buyer_balance`	Needed to confirm money transfers are credited/debited to the correct players.
052	test_game_decision_paths.py	`test_auction_property_leaves_it_unowned_when_everyone_passes`	Needed to cover auction branch behavior and bid resolution rules.
053	test_game_decision_paths.py	`test_handle_jail_turn_uses_a_jail_free_card`	Needed to verify jail entry/release/payment paths and related state changes.
054	test_game_decision_paths.py	`test_handle_jail_turn_can_decline_the_card_and_pay_instead`	Needed to verify jail entry/release/payment paths and related state changes.
055	test_game_decision_paths.py	`test_handle_jail_turn_waits_when_the_player_declines_to_leave`	Needed to verify jail entry/release/payment paths and related state changes.
056	test_game_decision_paths.py	`test_handle_jail_turn_forces_release_after_three_turns`	Needed to verify jail entry/release/payment paths and related state changes.

#	Test File	Test Function	Why Needed
057	test_game_decision_paths.py	`test_apply_card_handles_none_and_unknown_actions`	Needed to verify card destination/action resolution and its side effects.
058	test_game_decision_paths.py	`test_apply_card_pay_and_jail_paths`	Needed to verify jail entry/release/payment paths and related state changes.
059	test_game_decision_paths.py	`test_apply_card_birthday_and_move_to_go_cover_remaining_handlers`	Needed to verify card destination/action resolution and its side effects.
060	test_game_decision_paths.py	`test_collect_from_other_players_skips_people_who_cannot_afford_it`	Needed to validate a distinct white-box branch/edge path for correctness and regression safety.
061	test_game_decision_paths.py	`test_check_bankruptcy_leaves_solvent_players_in_the_game`	Needed to verify bankruptcy transitions are correct and players are not wrongly eliminated.
062	test_game_decision_paths.py	`test_check_bankruptcy_releases_properties_and_resets_index`	Needed to verify bankruptcy transitions are correct and players are not wrongly eliminated.
063	test_game_decision_paths.py	`test_check_bankruptcy_tolerates_players_already_removed_from_the_game`	Needed to verify bankruptcy transitions are correct and players are not wrongly eliminated.
064	test_game_decision_paths.py	`test_find_winner_returns_none_for_an_empty_game`	Needed to ensure winner/asset valuation logic is accurate and deterministic.
065	test_game_decision_paths.py	`test_run_prints_a_winner_when_one_player_remains`	Needed to ensure winner/asset valuation logic is accurate and deterministic.
066	test_game_decision_paths.py	`test_run_executes_the_loop_and_prints_standings`	Needed to validate a distinct white-box branch/edge path for correctness and regression safety.
067	test_game_decision_paths.py	`test_run_handles_an_empty_player_list`	Needed to validate a distinct white-box branch/edge path for correctness and regression safety.
068	test_game_decision_paths.py	`test_interactive_menu_dispatches_all_visible_options`	Needed to validate branch-heavy turn flow and state transition correctness.
069	test_game_decision_paths.py	`test_interactive_menu_skips_non_positive_loan_requests`	Needed to validate branch-heavy turn flow and state transition correctness.
070	test_game_decision_paths.py	`test_interactive_menu_handles_unaffordable_emergency_loans`	Needed to validate branch-heavy turn flow and state transition correctness.
071	test_game_decision_paths.py	`test_interactive_menu_ignores_invalid_choices_and_loops`	Needed to validate branch-heavy turn flow and state transition correctness.
072	test_game_decision_paths.py	`test_menu_mortgage_covers_empty_and_valid_selection`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
073	test_game_decision_paths.py	`test_menu_mortgage_ignores_invalid_selection`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
074	test_game_decision_paths.py	`test_menu_unmortgage_covers_empty_and_valid_selection`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
075	test_game_decision_paths.py	`test_menu_unmortgage_ignores_invalid_selection`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.

#	Test File	Test Function	Why Needed
076	test_game_decision_paths.py	`test_menu_trade_covers_missing_players_invalid_choices_and_success`	Needed to confirm money transfers are credited/debited to the correct players.
077	test_helpers_and_cli.py	`test_bank_zero_payout_and_insufficient_payout`	Needed to verify loan handling and bank ledger consistency.
078	test_helpers_and_cli.py	`test_bank_give_loan_ignores_non_positive_amounts`	Needed to verify loan handling and bank ledger consistency.
079	test_helpers_and_cli.py	`test_bank_summary_and_repr`	Needed to verify loan handling and bank ledger consistency.
080	test_helpers_and_cli.py	`test_board_reports_blank_and_special_tiles`	Needed to validate board/property invariants, ownership, and purchase rules.
081	test_helpers_and_cli.py	`test_board_purchasable_checks_cover_none_owned_and_mortgaged`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
082	test_helpers_and_cli.py	`test_board_repr_reflects_owned_property_count`	Needed to validate board/property invariants, ownership, and purchase rules.
083	test_helpers_and_cli.py	`test_card_deck_peek_len_repr_and_reshuffle`	Needed to verify card destination/action resolution and its side effects.
084	test_helpers_and_cli.py	`test_dice_reset_describe_and_repr`	Needed to ensure helper randomness/deck operations stay within valid bounds.
085	test_helpers_and_cli.py	`test_dice_roll_increments_the_doubles_streak`	Needed to ensure helper randomness/deck operations stay within valid bounds.
086	test_helpers_and_cli.py	`test_player_validation_and_display_helpers`	Needed to validate a distinct white-box branch/edge path for correctness and regression safety.
087	test_helpers_and_cli.py	`test_property_helpers_cover_mortgage_availability_and_repr`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
088	test_helpers_and_cli.py	`test_property_group_helpers_cover_owner_counts_size_and_repr`	Needed to validate board/property invariants, ownership, and purchase rules.
089	test_helpers_and_cli.py	`test_ui_print_helpers_cover_player_board_and_standings`	Needed to validate board/property invariants, ownership, and purchase rules.
090	test_helpers_and_cli.py	`test_ui_print_player_card_handles_players_without_properties`	Needed to verify card destination/action resolution and its side effects.
091	test_helpers_and_cli.py	`test_ui_input_helpers_cover_fallbacks_and_negative_confirmation`	Needed to validate startup and CLI robustness under invalid or EOF inputs.
092	test_helpers_and_cli.py	`test_get_player_names_trims_and_filters_empty_entries`	Needed to validate a distinct white-box branch/edge path for correctness and regression safety.
093	test_helpers_and_cli.py	`test_main_runs_the_game_when_setup_succeeds`	Needed to validate startup and CLI robustness under invalid or EOF inputs.
094	test_helpers_and_cli.py	`test_main_handles_keyboard_interrupt`	Needed to validate board/property invariants, ownership, and purchase rules.
095	test_helpers_and_cli.py	`test_main_handles_eof_during_initial_player_input`	Needed to validate startup and CLI robustness under invalid or EOF inputs.

#	Test File	Test Function	Why Needed
096	test_helpers_and_cli.py	`test_main_handles_setup_value_errors`	Needed to validate startup and CLI robustness under invalid or EOF inputs.
097	test_helpers_and_cli.py	`test_main_module_runs_from_the_script_entry_point`	Needed to validate startup and CLI robustness under invalid or EOF inputs.
098	test_jail_and_mortgage.py	`test_jail_fine_is_deducted_before_release`	Needed to verify jail entry/release/payment paths and related state changes.
099	test_jail_and_mortgage.py	`test_cannot_pay_voluntary_jail_fine_without_cash`	Needed to verify jail entry/release/payment paths and related state changes.
100	test_jail_and_mortgage.py	`test_unmortgage_fails_without_changing_state_when_cash_is_short`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
101	test_player_movement.py	`test_move_collects_salary_when_passing_go`	Needed to verify pass-Go salary behavior and movement economy rules.
102	test_player_movement.py	`test_move_collects_salary_when_landing_on_go`	Needed to verify pass-Go salary behavior and movement economy rules.
103	test_player_movement.py	`test_move_without_wrapping_does_not_award_salary`	Needed to validate a distinct white-box branch/edge path for correctness and regression safety.
104	test_purchase_guards.py	`test_buy_property_rejects_already_owned_property`	Needed to validate board/property invariants, ownership, and purchase rules.
105	test_purchase_guards.py	`test_buy_property_rejects_mortgaged_unowned_property`	Needed to validate mortgage state, payment guards, and prevent illegal state transitions.
106	test_setup.py	`test_game_rejects_invalid_player_counts`	Needed to validate a distinct white-box branch/edge path for correctness and regression safety.
107	test_setup.py	`test_game_keeps_cleaned_player_names`	Needed to validate a distinct white-box branch/edge path for correctness and regression safety.
108	test_transfers.py	`test_pay_rent_transfers_cash_to_owner`	Needed to confirm money transfers are credited/debited to the correct players.
109	test_transfers.py	`test_pay_rent_skips_mortgaged_properties`	Needed to confirm money transfers are credited/debited to the correct players.
110	test_transfers.py	`test_trade_moves_cash_and_property_to_the_right_players`	Needed to confirm money transfers are credited/debited to the correct players.
111	test_transfers.py	`test_trade_rejects_negative_cash_amounts`	Needed to confirm money transfers are credited/debited to the correct players.
112	test_turn_menu.py	`test_play_turn_opens_the_pre_roll_menu`	Needed to validate branch-heavy turn flow and state transition correctness.
113	test_turn_order.py	`test_bankruptcy_does_not_skip_the_next_player`	Needed to verify bankruptcy transitions are correct and players are not wrongly eliminated.

Total: 113 passing test cases | 17 failures all fixed | Final score: 10/10 after 6 iterations